Physics-Constrained Machine Learning

Concepts, Tools, and Algorithms

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Module 3: Estimation and Uncertainty Quantification

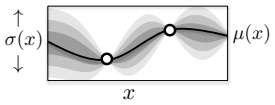
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Why Uncertainty Quantification?

- Physics-informed ML models are trained with limited and noisy data; their params and predictions are thus *uncertain*.
- Uncertainty quantification (UQ) tells us how much to trust a model and its predictions.
- UQ of physics-informed models is key for:
  - Reliability and robustness testing (e.g., will the process meet specifications?).
  - Experimental design (e.g., which experiment is most informative?).
  - Bayesian optimization (balance exploration and exploitation; Module 4).
  - Real-time optimization and control (e.g., stochastic and robust predictive control).
- UQ also helps compare models (e.g., how much physics to embed) and assess generalization.

Estimation and UQ: Overview

Gaussian Process (Non-Parametric)



$$y(x) \sim \mathcal{N}(\mu(x), \sigma^2(x))$$
$$y_s(x),\ s \in \mathcal{S}$$

NN Structures (Parametric)



$$y = \mathrm{PH}_\theta(x, \mathrm{ML}_\theta(x)) = \mathrm{PHML}_\theta(x)$$

Laplace Approximation

$$p(\theta|y) \approx \mathcal{N}(\theta^*, H_{\theta^*}^{-1})$$
$$\theta^* \in \operatorname*{argmax}_\theta \log p(y|\theta) + \log p(\theta)$$
$$y_s(x) = \mathrm{PHML}_{\theta_s}(x),\ s \in \mathcal{S}$$

Variational Inference

$$p(\theta|y) \approx q(\theta|\lambda)$$
$$\min_\lambda \frac{1}{S}\sum_{s\in\mathcal{S}} (\log q(\theta_s|\lambda) - \log p(\theta_s) - \log p(y|\theta_s))$$
$$y_s(x) = \mathrm{PHML}_{\theta_s}(x),\ s \in \mathcal{S}$$

- **Key:** GPs (easy UQ), NNs and hybrid models (complex UQ).

Gawlikowski, J., et al. (2023). A survey of uncertainty in deep neural networks. Artif. Intell. Rev., 56(Suppl 1), 1513-1589. doi:10.1007/s10462-023-10562-9

Models and Parameters

- We consider general models of the form:

$$y_\omega = m(\theta, \mathbf{x}_\omega) + \epsilon_\omega,\ \omega \in \mathcal{S}$$

- Model captures structural relationships between inputs, params, and outputs.
- This can be a physics model with physical params θ (e.g., reactor model, kinetic constants).
- It also covers NNs and hybrids of Modules 1-2, $m(\theta, x) = \mathrm{PHML}_\theta(x)$; $\theta \in \mathbb{R}^p$ holds physical params and NN weights/biases.
- We express model in compact form $\mathbf{y} = \mathbf{m}(\theta, \mathbf{X}) + \mathbf{e}$ and assume $\mathbf{e} \sim \mathcal{N}(0, \Sigma)$.

Maximum Likelihood Estimation

- We assume θ and $\mathbf{X}$ are deterministic and thus $\mathbf{y} \sim \mathcal{N}(\mathbf{m}(\theta), \Sigma)$.
- Maximum likelihood estimation (MLE) finds params that make the data most likely by minimizing $-\log f(\mathbf{y}|\theta)$:

$$\min_{\theta} \ \log\left((2\pi)^{S/2}|\Sigma|^{1/2}\right) + \frac{1}{2}(\mathbf{y}-\mathbf{m}(\theta))^T\Sigma^{-1}(\mathbf{y}-\mathbf{m}(\theta))$$

- If Σ is diagonal with entries $\Sigma_{\omega,\omega} = \sigma^2$, MLE reduces to sum of squared errors (SSE) minimization; this is the data loss $L_d(\theta)$ used to train NNs in Modules 1 and 2.
- Solution $\hat{\theta}$ must satisfy necessary conditions (score equations):

$$-\nabla_\theta \log f(\mathbf{y}|\theta) = -\mathcal{J}(\theta)^T\Sigma^{-1}(\mathbf{y}-\mathbf{m}(\theta)) = 0$$

where $\mathcal{J}(\theta) = \nabla_\theta \mathbf{m}(\theta)$ is the Jacobian of the model.

- Nonlinear score equations are solved numerically (e.g., Newton or gradient methods).

The Hessian Matrix

- To ensure that $\hat{\theta}$ minimizes $-\log f(\mathbf{y}|\theta)$, we must ensure $\mathcal{H}(\hat{\theta}) \succ 0$ holds.
- Here, $\mathcal{H}(\hat{\theta}) = -\nabla^2_\theta \log f(\mathbf{y}|\hat{\theta}) \in \mathbb{R}^{p\times p}$ is the Hessian (matrix of second derivatives):

$$\mathcal{H}(\theta)_{i,j} = -\frac{\partial^2 \log f(\mathbf{y}|\theta)}{\partial\theta_i \partial\theta_j}, \ i,j = 1,...,p.$$

- When residuals $\mathbf{y} - \mathbf{m}(\theta)$ are small, Hessian can be approximated as (Gauss-Newton):

$$\mathcal{H}(\hat{\theta}) \approx \mathcal{J}(\hat{\theta})^T\Sigma^{-1}\mathcal{J}(\hat{\theta}).$$

- Hessian measures information content of data (Fisher information):
  - Large eigenvalues: params are well-defined by data (sharp minimum).
  - Eigenvalues close to zero: params are ill-defined by data (flat minimum).

Prior Knowledge

- We often know roughly what param values should be (e.g., not negative or large).
- Prior knowledge helps when data cannot estimate params uniquely (zero Hessian eigenvalues) or reliably (near-zero eigenvalues).
- Prior knowledge enters as constraints (e.g., bounds, physics) or as a penalty (regularization):

$$\min_{\theta} \ -\log f(\mathbf{y}|\theta) + \kappa \cdot \rho(\theta)$$

- Penalty balances *prior knowledge* from $\rho(\theta)$ with *new knowledge* in data.

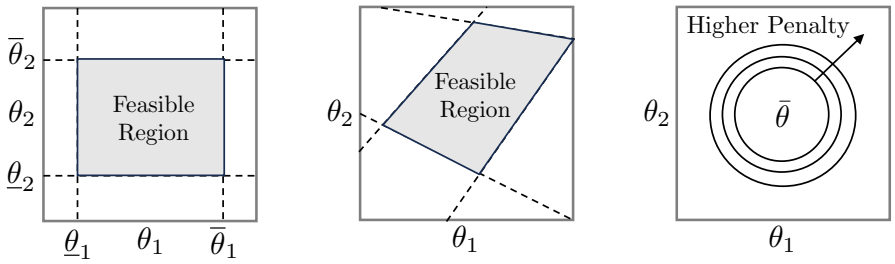


Figure: Feasible region defined by bounds (left) and general constraints (middle). Penalization induced by ridge penalty (right).

Constraints on Parameters

- Prior knowledge can be imposed as constraints on parameters; e.g., bounds or general (linear or nonlinear) constraints:

$$\min_{\theta} \ -\log f(\mathbf{y}|\theta) \quad \text{s.t.} \quad \underline{\theta} \le \theta \le \overline{\theta}, \quad \underline{\theta} \le \Pi\theta \le \overline{\theta}, \quad h(\theta) = 0$$

- Constraints define a *feasible region*: they restrict the estimation process, narrow down the search space, and eliminate spurious estimates that fit the data but have no physical meaning (e.g., negative rate constants).
- **Key insight:** this is directly analogous to physics-constrained ML. Physical constraints (e.g., mass balances) restrict what the ML model can predict:

$$\min_{\theta} \ -\log f(\mathbf{y}|\theta) \quad \text{s.t.} \quad \mathrm{PH}(\mathbf{x}, \mathbf{m}(\theta, \mathbf{x})) = 0$$

- In both cases, constraints (physics) better define estimation: they reduce degrees of freedom and help solvers avoid spurious solutions.

Example: Spurious Solutions PCML_Module3_Spurious_HougenWatson

- Hougen-Watson model $Y = (\theta_0 X_2 - X_3/\theta_4)/(1 + \theta_1 X_1 + \theta_2 X_2 + \theta_3 X_3)$, 13 observations.
- Different initial guesses give a physical estimate (SSE = 0.30) and a non-physical estimate with negative params (SSE = 0.38); both fit the data well.
- Bounds $\theta \geq 0$ (physical knowledge) eliminate the spurious solution.
- Analogy (Module 2): unconstrained NN fits membrane data but violates mass balances by up to 30 kg/h; physical constraints eliminate these spurious predictions.

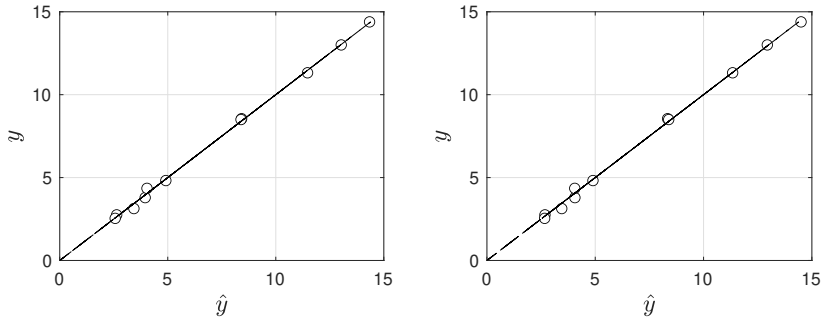


Figure: Fit of Hougen-Watson model with physical (left) and non-physical (right) parameters.

Bayesian Estimation

- Bayesian estimation unifies MLE and prior knowledge. Bayes' theorem states that:

$$f(\theta|\mathbf{y}) = \frac{f(\mathbf{y}|\theta)f(\theta)}{f(\mathbf{y})} \propto f(\mathbf{y}|\theta)f(\theta)$$

- $f(\theta|\mathbf{y})$ is the *posterior*: prob that params take value θ given data $\mathbf{y}$.
- $f(\theta)$ is the *prior*: prob that params take value θ before observing data.
- $f(\mathbf{y}|\theta)$ is the *likelihood* maximized in MLE.
- Maximum a posteriori (MAP) estimate maximizes probability of params given data:

$$\max_{\theta} \quad \log f(\mathbf{y}|\theta) + \log f(\theta)$$

- Prior term $\log f(\theta)$ naturally carries prior knowledge.

Bayesian Estimation: Prior as Regularization

- Consider model $\mathbf{y} = \mathbf{m}(\theta) + \mathbf{e}$ with $\mathbf{e} \sim \mathcal{N}(0, \Sigma)$ and prior $\theta \sim \mathcal{N}(\bar{\theta}, \Sigma_\theta)$.
- After dropping constant terms, MAP estimation solves:

$$\min_{\theta} \quad \underbrace{\frac{1}{2}(\mathbf{y} - \mathbf{m}(\theta))^T \Sigma^{-1} (\mathbf{y} - \mathbf{m}(\theta))}_{-\log f(\mathbf{y}|\theta)} + \underbrace{\frac{1}{2}(\theta - \bar{\theta})^T \Sigma_\theta^{-1} (\theta - \bar{\theta})}_{-\log f(\theta)}$$

- The prior is a *regularizer*; e.g., $\bar{\theta} = 0$ and $\Sigma_\theta = \alpha^{-1} I$ gives the ridge penalty $\kappa\,\rho(\theta)$ used to train NNs ($\kappa = \alpha/2$, $\rho(\theta) = \|\theta\|^2$).
- Bayesian estimation strikes a balance between what we knew about θ and the new knowledge in data $\mathbf{y}$.
- Prior knowledge *reduces model ambiguity*; trust in the prior is set by Σ_θ.

Example: Bayesian Estimation PCML_Module3_Bayes_LinearModel

- Linear model $Y = \theta X + \epsilon$, one observation $(x = 1, y = 2)$, prior $\theta \sim \mathcal{N}(0, 1)$.
- Prior says $\theta = 0$; data (MLE) say $\theta = 2$; posterior blends both: $\theta = 1$.

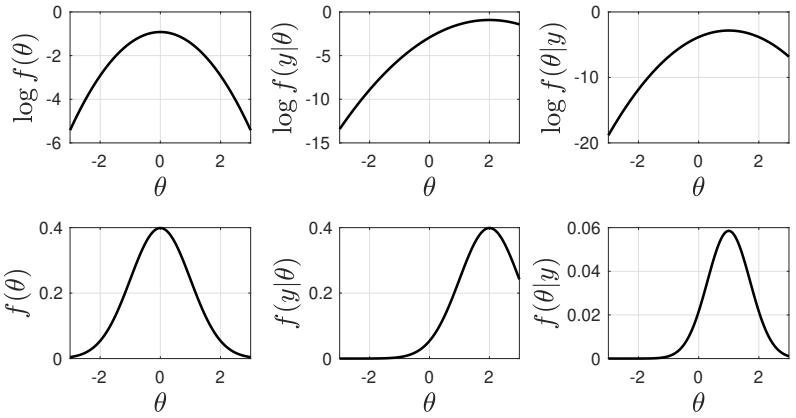


Figure: Prior $f(\theta)$, likelihood $f(y|\theta)$, and posterior $f(\theta|y)$ (top: log scale).

Estimation for Physics-Informed ML Models

- Physics-informed ML models follow the same logic (MLE or Bayesian), but:
  - The number of params is much larger (NN weights and biases + physical params).
  - NNs are highly nonlinear; estimation is nonconvex with many (nearly) equivalent minima.
  - The Hessian is often singular or indefinite (params ill-defined by data).
  - Solution depends on initialization; first-order methods (Adam) are typically used.
- Priors (regularization) and physics (constraints) make estimation well-posed.
- Estimation of physics-informed models is harder than training a pure NN when physics is nonlinear or involves differential equations (repeated simulations).

Recall: Membrane Example PCML_Module3_UQ_Membrane

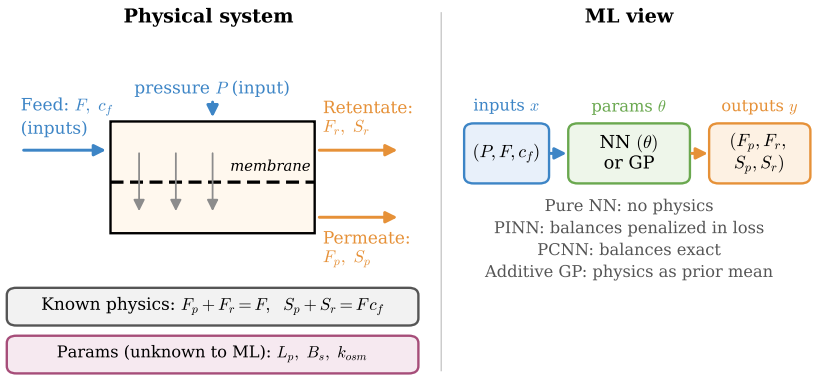


Figure: Membrane separator as a physical system (left) and as seen by ML models (right).

- We estimate the params θ of the PCNN of Module 2 from 10 noisy experiments.
- Then we quantify how uncertain θ and the predicted flows are (Laplace, VI).

Example: MLE vs. MAP for Membrane PCML_Module3_UQ_Membrane

- PCNN of Module 2 (membrane, 10 experiments) written as $m(\theta, \mathbf{x})$ with $\theta \in \mathbb{R}^{50}$; prior $\theta \sim \mathcal{N}(0, \alpha^{-1} I)$ with $\alpha = 10$.
- MLE from 3 initializations: F_p RMSE 3.5 to 5.1 kg/h; Hessian is indefinite (ill-posed).
- MAP from 3 initializations: RMSE of F_p 1.9 kg/h for all; Hessian is positive definite.

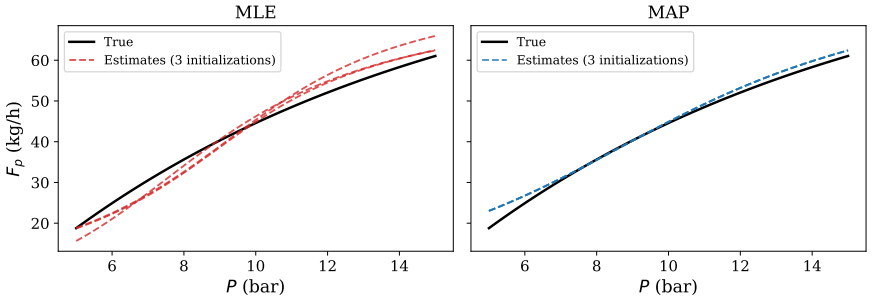


Figure: Permeate flow vs. pressure ($F = 100$ kg/h, $c_f = 0.02$) for estimates obtained from 3 initializations.

Uncertainty Quantification

- Estimates $\hat{\theta}$ depend on noisy data; their uncertainty is captured by the posterior $f(\theta|\mathbf{y})$.
- Uncertainty of params propagates to uncertainty of model predictions $y(x) = m(\theta, x)$.
- For linear models, posterior is Gaussian and can be obtained exactly.
- For nonlinear models (NNs, hybrids), posterior must be approximated:
  - Laplace approximation (uses Hessian)
  - Variational inference (uses optimization)
  - Sampling (e.g., Monte Carlo, bootstrapping, ensembles; expensive)

Laplace Approximation

- Laplace approximation fits a Gaussian to the posterior around the MAP estimate:

$$f(\theta|\mathbf{y}) \approx \mathcal{N}\left(\hat{\theta},\ \mathcal{H}(\hat{\theta})^{-1}\right), \qquad \mathcal{H}(\hat{\theta}) = -\nabla^2_\theta \left[\log f(\mathbf{y}|\theta) + \log f(\theta)\right]_{\theta=\hat{\theta}}$$

- Covariance of params is the *inverse of the Hessian*: $\mathrm{Cov}[\hat{\theta}] = \mathcal{H}(\hat{\theta})^{-1}$.

- With Gauss-Newton approximation and Gaussian prior:

$$\mathrm{Cov}[\hat{\theta}] = (\mathcal{J}(\hat{\theta})^T \Sigma^{-1} \mathcal{J}(\hat{\theta}) + \Sigma_\theta^{-1})^{-1}.$$

  Formula includes contribution of likelihood (data) and of prior.

- Exact for linear models; accurate if nonlinearity is mild (local approximation).

Laplace Approximation: Uncertainty of Predictions

- **Sampling:** draw $\theta_s \sim \mathcal{N}(\hat{\theta}, \mathcal{H}(\hat{\theta})^{-1})$, propagate $y_s(x) = m(\theta_s, x),\ s = 1, ..., N_s$; mean and quantiles give prediction and bands.
- **Linearization:** linearize model around $\hat{\theta}$ (closure of Gaussians under linear transformations):

$$\mathbb{V}[y(x)] \approx \nabla_\theta m(\hat{\theta}, x)^T\, \mathcal{H}(\hat{\theta})^{-1}\, \nabla_\theta m(\hat{\theta}, x)$$

- Linearization is cheap and robust; sampling far from $\hat{\theta}$ in highly nonlinear models (NNs) can give unreliable predictions.

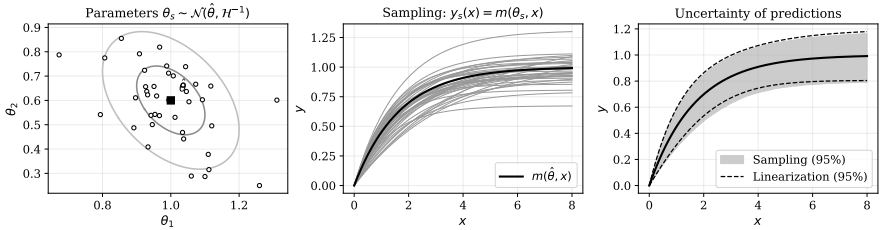


Figure: Model $y = \theta_1(1 - e^{-\theta_2 x})$: posterior samples (left), sampled predictions (middle), and 95% bands (right).

Limitations of the Laplace Approximation

- Laplace approximation requires the Hessian $\mathcal{H}(\hat{\theta}) \in \mathbb{R}^{p \times p}$:
  - Entries grow quadratically: a NN with 10^6 params has a Hessian with 10^{12} entries.
  - Inverting (factorizing) Hessian scales as $\mathcal{O}(p^3)$.
  - Hessian might be indefinite (nonconvex problem) or singular.
  - Many software packages do not provide Hessians.
- For neural ODEs, the Hessian requires second derivatives through simulations.
- Hessian might thus be impossible to get in complex models.
- This motivates UQ techniques that only need gradients, such as *variational inference*.

Example: Laplace Approximation for Membrane PCML_Module3_UQ_Membrane

- Hessian of $-\log f(\mathbf{y}|\theta) - \log f(\theta)$ at MAP estimate is 50×50; PyTorch computes it with automatic differentiation in 0.01 s.
- Param standard deviations range from 0.1 to 0.8 (some params poorly defined by data).
- Uncertainty of predictions obtained by linearization; 95% intervals contain the true F_p for 78% of test inputs (average width 4.5 kg/h).
- For this small model, Laplace approximation is easy; for large NNs and neural ODEs, it is not.

Variational Inference

- Idea: approximate posterior $f(\theta|\mathbf{y})$ with a simple density $q(\theta|\lambda)$ easy to sample.
- λ are *variational params*; e.g., mean-field Gaussian (independent components):

$$q(\theta|\lambda) = \prod_{i=1}^{p} \mathcal{N}(\theta_i \,|\, \mu_i, \sigma_i^2), \qquad \lambda = (\mu, \rho), \qquad \sigma_i = e^{\rho_i}$$

- We find λ that makes $q(\theta|\lambda)$ as close as possible to posterior, measured by Kullback-Leibler (KL) divergence:

$$\min_{\lambda} \ \mathrm{KL}\left(q(\theta|\lambda) \,\|\, f(\theta|\mathbf{y})\right) = \mathbb{E}_q\left[\log q(\theta|\lambda) - \log f(\theta|\mathbf{y})\right]$$

- VI casts UQ as *optimization* (instead of Hessian calculation or sampling).

Blei, D. M., Kucukelbir, A., and McAuliffe, J. D. (2017). Variational inference: A review for statisticians. JASA, 112(518), 859-877. doi:10.1080/01621459.2017.1285773

Variational Inference: Evidence Lower Bound

- Using Bayes' theorem, $\log f(\theta|\mathbf{y}) = \log f(\mathbf{y}|\theta) + \log f(\theta) - \log f(\mathbf{y})$, and thus:

$$\mathrm{KL}\left(q \,\|\, f(\theta|\mathbf{y})\right) = -\underbrace{\left(\mathbb{E}_q\left[\log f(\mathbf{y}|\theta)\right] - \mathrm{KL}\left(q(\theta|\lambda) \,\|\, f(\theta)\right)\right)}_{\mathrm{ELBO}(\lambda)} + \log f(\mathbf{y})$$

- $\log f(\mathbf{y})$ does not depend on λ; minimizing KL is equivalent to maximizing the *evidence lower bound* (ELBO):

$$\max_{\lambda} \ \mathrm{ELBO}(\lambda) = \underbrace{\mathbb{E}_q\left[\log f(\mathbf{y}|\theta)\right]}_{\text{fit data}} - \underbrace{\mathrm{KL}\left(q(\theta|\lambda) \,\|\, f(\theta)\right)}_{\text{stay close to prior}}$$

- For Gaussian q and prior, KL term has closed form; expectation is estimated by sampling.

Variational Inference: Reparameterization

- Expectation estimated by Monte Carlo with the *reparameterization trick*:

$$\theta_s = \mu + \sigma \odot \varepsilon_s, \qquad \varepsilon_s \sim \mathcal{N}(0, I), \qquad s = 1, ..., N_s$$

- This gives the estimate:

$$\mathrm{ELBO}(\lambda) \approx \frac{1}{N_s} \sum_{s=1}^{N_s} \log f(\mathbf{y}|\theta_s) - \mathrm{KL}\left(q(\theta|\lambda) \,\|\, f(\theta)\right)$$

- Samples θ_s are differentiable functions of $\lambda = (\mu, \rho)$; thus, $\nabla_\lambda \mathrm{ELBO}$ can be computed by back propagation (through the model and the samples).
- Only a handful of samples N_s (e.g., 1 to 10) are needed per iteration.

Kingma, D. P., and Welling, M. (2014). Auto-encoding variational Bayes. ICLR. doi:10.48550/arXiv.1312.6114

Variational Inference: Gradient-Based Solution

- VI is solved with the same gradient-based tools used to train NNs:

Initialize $\lambda_0 = (\mu_0, \rho_0)$ (e.g., $\mu_0 = \hat{\theta}_{MAP}$)
for $\ell = 0, 1, 2, ...$ **do**
  Draw $\varepsilon_s \sim \mathcal{N}(0, I)$ and set $\theta_s = \mu_\ell + \sigma_\ell \odot \varepsilon_s$, $s = 1, ..., N_s$
  Estimate $\mathrm{ELBO}(\lambda_\ell)$ and compute $\nabla_\lambda \mathrm{ELBO}(\lambda_\ell)$ by back propagation
  Update $\lambda_{\ell+1}$ with a stochastic gradient step (e.g., Adam)
end for

- Gradient is noisy (Monte Carlo samples); Adam handles this well.
- VI has $2p$ variational params and *needs no Hessian*; it scales to large NNs and neural ODEs.
- Predictions: sample $\theta_s \sim q(\theta|\lambda^*)$ and propagate $y_s(x) = m(\theta_s, x)$.
- Mean-field q ignores correlations between params and can underestimate uncertainty.

Example: VI for Membrane PCML_Module3_UQ_Membrane

- Mean-field Gaussian $q(\theta|\mu,\rho)$ with 100 variational params; 8 samples per iteration; Adam (8000 iterations, 9 s).
- ELBO estimate is noisy (Monte Carlo) but its average converges.
- 95% intervals contain the true F_p for 85% of test inputs (average width 5.0 kg/h).

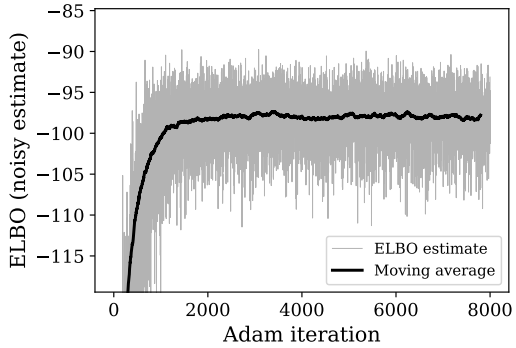


Figure: ELBO estimates during Adam iterations.

Example: Laplace vs. VI for Membrane PCML_Module3_UQ_Membrane

- Both methods give similar mean predictions and uncertainty bands (coverage of 78% for Laplace and 85% for VI).
- Both are approximations and tend to be somewhat overconfident (Laplace is local; mean-field VI ignores correlations).
- Laplace requires the Hessian; VI only requires gradients.

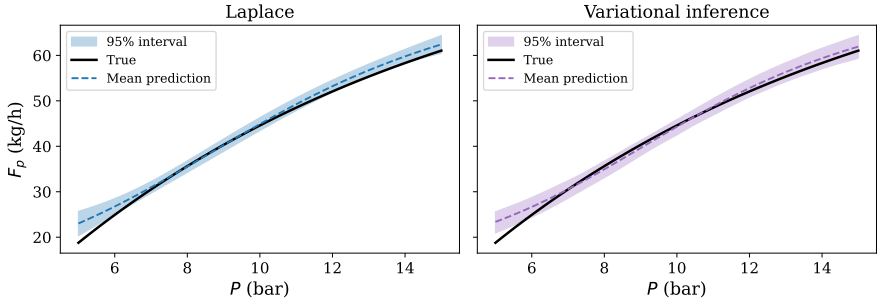


Figure: Permeate flow vs. pressure with 95% intervals for Laplace (left) and VI (right).

Example: When Laplace Breaks PCML_Module3_UQ_Membrane

- Strong nonlinearity: $y = \theta^3 x + \epsilon$ (vanishing gradient near $\theta = 0$, as in saturated neurons); 10 observations, $\theta_{true} = 0.15$; exact posterior computed on a grid.
- Exact posterior is far from Gaussian (flat and skewed); Laplace uses only local curvature at the mode: mean 0.26 vs. exact 0.07, and its prediction band is shifted and too wide.
- VI optimizes over the whole posterior and stays consistent with exact posterior (mean 0.06, std 0.16 vs. exact 0.19) and with exact prediction band.

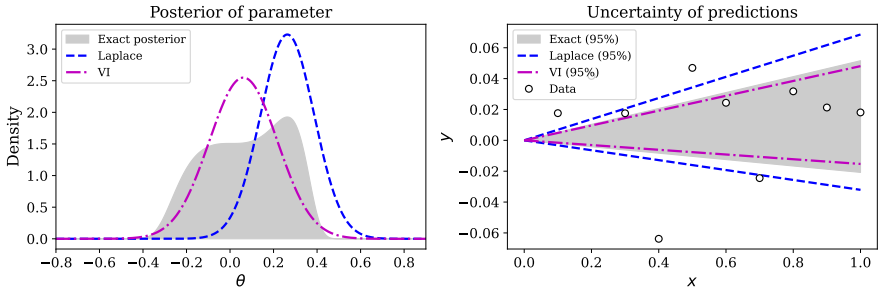


Figure: Exact posterior, Laplace, and VI for the parameter (left) and 95% bands for predictions (right).

Example: VI for Microbial Communities

- VI estimates params and uncertainty of physics, hybrid (NSM), and ML (eNODE) models.
- Uncertainty bands reveal how models generalize: physics model misses key trends; hybrid models capture the dynamics with consistent uncertainty.
- Hyperparams (prior precision, noise) estimated by expectation maximization.

Thompson, J., Connors, B. M., Zavala, V. M., and Venturelli, O. S. (2026). Physics-constrained neural ordinary differential equation models to discover and predict microbial community dynamics. PNAS, 123(13), e2517661123. doi:10.1073/pnas.2517661123

Example: VI for Microbial Communities

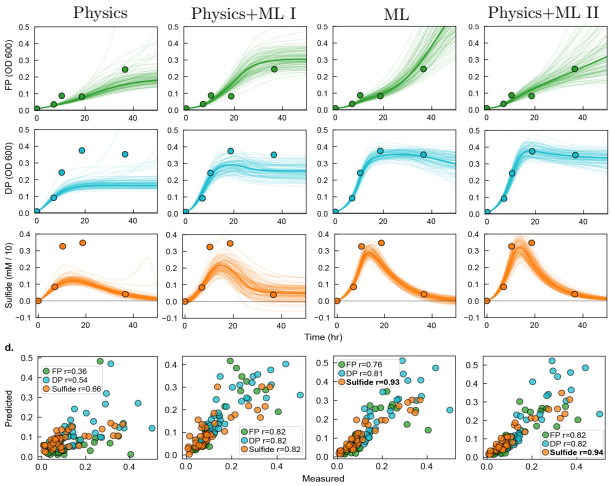


Figure: Predictions (with uncertainty) of physics, hybrid, and ML models.

Thompson, J., Connors, B. M., Zavala, V. M., and Venturelli, O. S. (2026). Physics-constrained neural ordinary differential equation models to discover and predict microbial community dynamics. PNAS, 123(13), e2517661123. doi:10.1073/pnas.2517661123

Gaussian Processes: Built-In Uncertainty

- GP models are *inherently probabilistic*: prediction at a new input x is Gaussian,

$$y(x) \sim \mathcal{N}(\mu(x), \sigma^2(x))$$

  with $\mu(x)$ and $\sigma^2(x)$ given in closed form (Module 1).

- Predictive uncertainty comes directly from the model (no Hessian or VI needed).
- Physics-informed GPs (e.g., physics model as prior mean) inherit this property.
- One can also draw samples $y_s(x)$, $s = 1, ..., N_s$, from the GP.
- In many applications, however, we need the uncertainty of a *function* of GP predictions; e.g., product purity, cost, or performance $f(y(x))$.

Propagating Uncertainty through Functions

- Performance $f(y(x))$ where $y(x) \sim \mathcal{N}(\mu_y(x), \Sigma_y(x))$ is given by GPs.
- We need mean $m_f(x)$ and uncertainty $\sigma_f(x)$ of performance.
- If f is linear, $f(y) = a^T y + b$, closure of Gaussians under linear transformations gives:

$$m_f(x) = a^T \mu_y(x) + b, \qquad \sigma_f^2(x) = a^T \Sigma_y(x)\, a$$

- If f is nonlinear (most cases), no closed form exists; we use *sampling* or *linearization*.
- This is necessary to apply techniques such as Bayesian optimization (BO) when performance is a function of GP models (physics-informed BO; Module 4), as well as for reliability analysis and stochastic control.

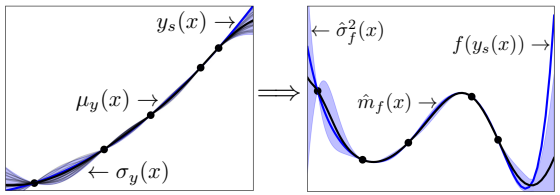
Propagation by Sampling

- Sample GP $y_s(x)$, $s = 1, ..., N_s$, propagate through f, and compute statistics:

$$\hat{m}_f(x) = \frac{1}{N_s} \sum_{s=1}^{N_s} f(y_s(x)), \qquad \hat{\sigma}_f^2(x) = \frac{1}{N_s - 1} \sum_{s=1}^{N_s} \left(f(y_s(x)) - \hat{m}_f(x)\right)^2$$

- Convenient and general (any f), but computationally expensive: many evaluations of f at every input x (and at every BO iteration).

Sample GP: $y_s(x), s \in \mathcal{S}$ Propagate: $f(y_s(x)), s \in \mathcal{S}$



Mean Performance

$$\hat{m}_f(x) = \frac{1}{S} \sum_{s \in \mathcal{S}} f(y_s(x))$$

Uncertainty Performance

$$\hat{\sigma}_f^2(x) = \frac{1}{S-1} \sum_{s \in \mathcal{S}} \left(f(y_s(x)) - \hat{m}_f(x)\right)^2$$

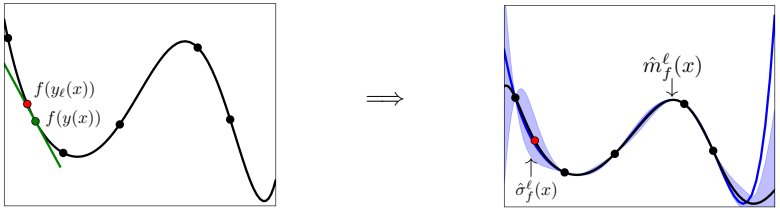
Propagation by Linearization

- Linearize f around a reference point $y_\ell(x)$ (e.g., GP mean):

$$f(y(x)) \approx f(y_\ell(x)) + a_\ell^T (y(x) - y_\ell(x)), \qquad a_\ell = \nabla_y f(y_\ell(x))$$

- Linear transformation of a Gaussian is Gaussian; thus $m_f(x) \approx f(y_\ell(x)) + a_\ell^T(\mu_y(x) - y_\ell(x))$ and $\sigma_f^2(x) \approx a_\ell^T \Sigma_y(x)\, a_\ell$.
- Analytical (one evaluation of f and gradient); Gaussianity preserved near reference point (Laplace); linearization updated adaptively (BOIS).

Linearization of $f(y(x))$ | Propagation of GP $y(x)$



$$f(y(x)) \approx f(y_\ell(x)) + \underbrace{\nabla_y f(y_\ell(x))^T}_{a_\ell}(y(x) - y_\ell(x))$$

$$\hat{m}_f^\ell(x) = a_\ell^T \mu_y(x) + b_\ell$$
$$\hat{\sigma}_f^\ell(x) = a_\ell^T \sigma_y(x) a_\ell$$

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Example: GP Uncertainty Propagation for Membrane PCML_Module3_UQ_Membrane

- GPs predict retentate flows F_r and S_r with uncertainty (10 experiments).
- We propagate uncertainty to retentate purity $c_r = f(F_r, S_r) = S_r/F_r$ (nonlinear) with gradient $\nabla f = (-S_r/F_r^2,\ 1/F_r)$.
- Sampling (2000 samples) and linearization give almost identical intervals (differences below 0.15 g/kg); linearization is about 60 times faster.

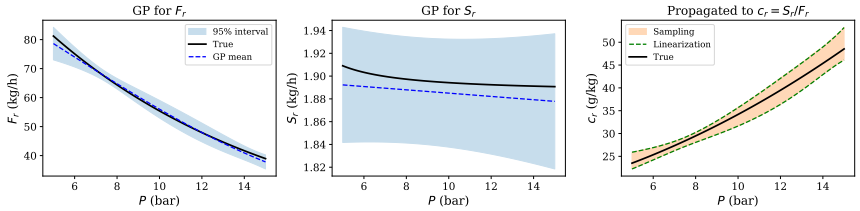


Figure: GP predictions of F_r and S_r (95% intervals) and propagated uncertainty of purity c_r by sampling and linearization.

Summary of UQ Techniques

Technique	Requires	Cost	Comments
Laplace	Hessian	p^2 memory, $\mathcal{O}(p^3)$	Local Gaussian
VI	Gradients	$2p$ params, Adam	Scalable; mean-field
Sampling/bootstrap	Repeated estimation	Many trainings	Simple, expensive
Ensembles	Repeated training	Many trainings	Initialization spread
GPs	Kernel matrix	$\mathcal{O}(S^3)$	Built-in UQ

- Physics-informed models need scalable UQ that respects physics; VI is promising.

Key Takeaways

- UQ of physics-informed models is key for reliability testing, experimental design, optimization, and control.
- Estimation of physics-informed ML models follows the logic of classical estimation (MLE, Bayesian), but problems are larger and highly nonlinear; priors make them well-posed.
- Laplace approximation quantifies uncertainty using the Hessian (inverse covariance of params), which might be impossible to obtain for complex models.
- Variational inference casts UQ as an optimization problem (maximize ELBO) that is solved with gradient-based methods (Adam), only requires gradients, and is more robust than Laplace for strongly nonlinear models.
- GPs are inherently probabilistic; their uncertainty is propagated through functions by sampling or (cheaper) linearization, as needed in Bayesian optimization.
- In the membrane example, MAP estimation was consistent where MLE was not, and Laplace and VI gave similar uncertainty bands.

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